NETWORK DEVICES AND HOME NETWORK

Assignment

Name: _____ Class: _____

Date: _____

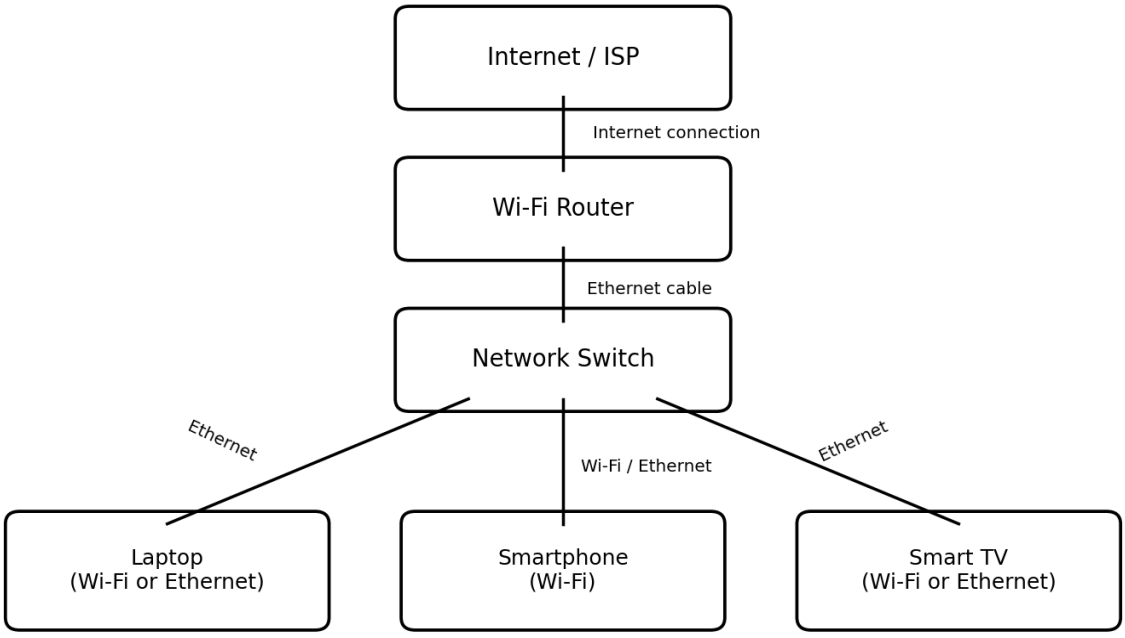
TASK 1

QUESTION

Draw and label a simple home network diagram showing a router, switch, and three devices (like a laptop, smartphone, and smart TV), indicating how each device connects to the network.

ANSWER

The diagram shows the internet connection coming to the Wi-Fi router. The router connects to the network switch using an Ethernet cable. The laptop, smartphone, and smart TV connect to the home network through the switch or Wi-Fi, depending on their connection method.



Note: In a typical home setup, the router may also have a built-in switch and Wi-Fi access point. The separate switch is shown for learning purposes.

TASK 2

QUESTION

Create a comparison table listing at least four network devices (router, switch, hub, access point) and describe one real-world use case for each, referencing apps or services used daily.

ANSWER

Network Device	Main Function	Real-World Use Case
Router	Connects a local network to the internet and directs traffic between networks.	When ordering food on Zomato or watching YouTube, the router forwards data between the home network and the internet.
Switch	Connects wired devices in the same LAN and sends data to the correct device.	A gaming café connects several gaming PCs to a switch for LAN gaming and local file transfers.
Hub	Connects Ethernet devices and sends incoming data to all ports.	An older training lab may use a hub to demonstrate shared traffic; switches are preferred in modern networks.
Access Point	Provides Wi-Fi so wireless devices can join a wired network.	A phone or smart TV connects over Wi-Fi to stream YouTube or use apps without an Ethernet cable.

Summary: A router connects networks, a switch connects wired devices in a LAN, a hub repeats data to all ports, and an access point provides wireless access.

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TASK 3

QUESTION

Write a short scenario explaining how data travels from your phone to the internet when you order food on Zomato, mentioning the role of each network device involved (router, switch, access point).

ANSWER

When I open the Zomato app on my smartphone and place a food order, the phone sends the order data through Wi-Fi to the wireless access point. In many home routers, the access point is built into the router; in a setup with separate devices, the access point passes the data through the wired network to a switch. The switch forwards the data toward the router, and the router sends it through the internet service provider to Zomato's servers. Zomato processes the order and sends a response back through the internet, router, switch (if used), and access point to my phone. The app then displays the order confirmation and tracking information.

Important note: In a typical home, the router, switch, and access point may be combined in one device, so a separate switch is not always involved.

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TASK 4

QUESTION

You have a Wi-Fi router, but your gaming PC only has an Ethernet port and is far from the router. Suggest two different network devices you could use to connect your PC to the internet and explain how each would work.

ANSWER

Option 1: Wi-Fi Extender with an Ethernet Port

Place a Wi-Fi extender where it receives a strong signal from the router. Connect the gaming PC to the extender's Ethernet port using an Ethernet cable. The extender receives the router's Wi-Fi signal and provides the connection to the PC through the cable. Performance depends on signal quality and placement.

Option 2: Powerline Adapter Kit

Plug one powerline adapter into a wall socket near the router and connect it to the router using an Ethernet cable. Plug the second adapter into a wall socket near the gaming PC and connect the PC to it using another Ethernet cable. The adapters carry network data over the home's electrical wiring. Actual speed and stability depend on the wiring; plug adapters directly into wall sockets when possible.

Note: A basic Ethernet switch alone will not solve the distance problem unless an Ethernet cable can be run from the router to the switch.

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